

21 VOID TECHNOLOGIES

ADMINISTRATOR & DEPLOYMENT MANUAL

Fusing Technology With Market Intelligence • Official System Documentation

1. Technical Architecture & Technology Stack

21 Void Technologies is built with an offline-first, container-ready Django 5 architecture.

- Backend: Python 3.10+, Django 5.2, strictly 100% Function-Based Views (FBVs).
- Input Validation: Zero Django Forms framework. All requests are handled via raw HTML inputs and validated by dedicated validation routines in `core/validators.py`.
- Frontend: Server-rendered templates, precompiled Tailwind CSS, self-hosted Inter fonts, FontAwesome 6 icons, jQuery 3.7.1 (sole AJAX client), SweetAlert2 (sole notification dialog system), and Chart.js 4.4.
- Database: SQLite (Local development) and PostgreSQL 16 (Docker environments).

2. Three Deployment Run Modes

The system includes a single launch script (`launch.bat` on Windows, `launch.sh` on Linux/macOS) supporting three isolated run modes:

Mode	Command	Database	Stack Architecture
Local Venv	<code>launch.bat local</code>	SQLite	Python venv + Django Dev Server
Docker Dev	<code>launch.bat docker_dev</code>	PostgreSQL 16	Docker Compose with Live Reload
Docker Prod	<code>launch.bat docker_prod</code>	PostgreSQL 16	Gunicorn (3 workers) + Nginx (port 80)

- Auto-Launch: In local and Docker prod modes, the launch script automatically opens the default web browser to the homepage upon startup.

3. User Roles & Role-Based Access Control (RBAC)

User permissions are enforced through the `role_required` decorator and custom session timeouts:

- Administrator (ADMIN): Full administrative rights, user account creation, system settings modifications, and sale voiding/deletion. Session timeout: 1 hour.
- Manager (MANAGER): Can add/edit products, record sales, access financial reports and view settings. Session timeout: 8 hours.
- Staff (STAFF): Operational point-of-sale checkout and inventory browsing. Restricted from sensitive deletions. Session timeout: 8 hours.

4. Company Settings & Receipt Customization

Administrators can configure company branding and receipt settings at /core/settings/:

- Company Name, Tagline, Phone, Email, Physical Address
- Currency Symbol (default: \$) and Currency Code (default: USD)
- Custom Receipt Header Titles & Return Policy / Guarantee Footer Notes

5. Automated Nightly Backups & Recovery

The system provides an automated backup utility (backup.py):

- SQLite Mode: Takes an atomic file-level copy of db.sqlite3 into the backups/ directory with timestamped filenames (e.g. db_backup_YYYYMMDD_HHMMSS.sqlite3).
- PostgreSQL Mode: Issues automated pg_dump commands to archive database tables into timestamped SQL dumps.
- Restoring Backups: For SQLite, simply replace db.sqlite3 with the backup copy. For PostgreSQL, execute pg_restore.

6. Non-Negotiable Constraints Compliance Verification

- Zero django.forms imports verified across all codebase files.
- Zero generic Class-Based Views (CBVs) verified.
- Zero fetch(), axios, or HTMX verified — jQuery \$.ajax is the sole client.
- Zero keyboard shortcuts or hotkeys anywhere — 100% mouse and touch driven.
- 100% Offline asset isolation verified — zero external CDN or remote API calls.